

Cluster-Prime Inside–Outside EM: Compute Scaling, Rank-4+ Tensor Contractions, and a Berkovich–Hodge Formulation

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Abstract

We analyze the computational scaling of cluster-prime Inside–Outside Expectation–Maximization (EM) for hierarchical tree grammars with $N = 10^6$ tokens and depth $L = 16$. We show that the full-symbol Inside recursion costs $O(NLV^3)$ operations and implodes at $V = 1024$, while the cluster-prime compression at $K = 16$ costs $O(NLK^3)$ and is tractable on a single GPU. We compare this against QKV Transformer training at matched scale, showing that the Transformer baseline is $10^6\times$ more expensive *and* architecturally incapable of detecting the hierarchy. We then generalize the framework to rank-4 and higher rule tensors (ternary and m -ary grammars), showing that the cluster-prime compression buys four to five extra tensor orders of headroom. Finally, we recast the scheme in the language of Berkovich spaces and Hodge filtrations, identifying the resolution radius r as the amortization knob that keeps cluster-prime discovery tractable.

1 Setup and Notation

Based on `estimator2.py`

Let V denote the symbol alphabet size, N the number of tokens, L the maximum tree depth, and K the number of cluster primes per node. We write $P_l(p \rightarrow x, y) \in \mathbb{R}^{V \times V \times V}$ for the rank-3 rule tensor at level l , and $\mathcal{P}_l \in \mathbb{R}^{V \times V \times V}$ for its full array.

The Inside recursion of Eq. (2) in the main paper is

$$T_l(b, n, p) = \sum_{x \in V} \sum_{y \in V} P_l(p \rightarrow x, y) \cdot T_{l-1}(b, 2n, x) \cdot T_{l-1}(b, 2n+1, y). \quad (1)$$

The Outside recursion has the same asymptotic cost.

2 Compute Analysis: Full-Symbol Inside–Outside

2.1 Per-node cost

For each parent $p \in V$ and each internal node (b, n) , evaluating (1) requires

$$\underbrace{V^2}_{\text{child pairs}} \times \underbrace{1}_{\text{rule lookup}} \times \underbrace{2}_{\text{multiplications}} = 2V^2 \quad (2)$$

operations. Summing over all V parents gives

$$C_{\text{node}}^{\text{full}} = 2V^3 \approx 2 \times 10^9 \quad \text{for } V = 1024. \quad (3)$$

2.2 Total cost

The number of internal nodes in a binary tree over N tokens is $N - 1 \approx N$. Over L levels:

$$C_{\text{Inside}}^{\text{full}} = NL \cdot 2V^3 = 2NLV^3. \quad (4)$$

With $N = 10^6$, $L = 16$, $V = 1024$:

$$C_{\text{Inside}}^{\text{full}} \approx 3.4 \times 10^{16} \text{ operations.} \quad (5)$$

The Outside recursion doubles this, and the M-step re-estimation adds another factor, giving

$$C_{\text{sweep}}^{\text{full}} \approx 4 \cdot NLV^3 \approx 6.9 \times 10^{16}. \quad (6)$$

Over $S = 50$ EM sweeps:

$$C_{\text{total}}^{\text{full}} \approx 3.4 \times 10^{18} \text{ operations.} \quad (7)$$

2.3 Memory

The rule tensor alone requires

$$V^3 \cdot 4 \text{ bytes} = 4 \text{ GB per level,} \quad (8)$$

or 64 GB across $L = 16$ levels. The Inside table requires

$$NLV \cdot 4 \text{ bytes} = 10^6 \cdot 16 \cdot 1024 \cdot 4 \approx 64 \text{ GB.} \quad (9)$$

Total memory exceeds 128 GB before any auxiliary storage.

Proposition 1 (Full-symbol implosion). *At $V = 1024$, $N = 10^6$, $L = 16$, the full-symbol Inside–Outside EM requires $\sim 10^{18}$ operations and > 128 GB of memory. It is not feasible on current hardware.*

3 Compute Analysis: Cluster-Prime Inside–Outside

3.1 Compressed recursion

Replace the symbol alphabet V with a per-node cluster set $C(p) = \{c_1, \dots, c_{K_p}\}$ with $K_p \leq K$. The compressed Inside recursion is

$$T_l(b, n, p) \approx \sum_{i=1}^{K_p} \sum_{j=1}^{K_p} \tilde{P}_l(p \rightarrow c_i, c_j) \cdot \tilde{T}_{l-1}(b, 2n, c_i) \cdot \tilde{T}_{l-1}(b, 2n+1, c_j), \quad (10)$$

where

$$\tilde{T}_{l-1}(b, \cdot, c_i) = \sum_{x \in c_i} T_{l-1}(b, \cdot, x) \quad (11)$$

is the cluster-marginalized Inside score.

3.2 Per-node cost

$$C_{\text{node}}^{\text{cluster}} = 2K^3. \quad (12)$$

Total per sweep:

$$C_{\text{sweep}}^{\text{cluster}} \approx 4NLK^3. \quad (13)$$

K	K^3	Ops per sweep	Total (50 sweeps)
2	8	5.1×10^8	2.6×10^{10}
4	64	4.1×10^9	2.0×10^{11}
8	512	3.3×10^{10}	1.6×10^{12}
16	4096	2.6×10^{11}	1.3×10^{13}
32	32768	2.1×10^{12}	1.0×10^{14}
64	262144	1.7×10^{13}	8.4×10^{14}
128	2.1×10^6	1.3×10^{14}	6.7×10^{15}
256	1.7×10^7	1.1×10^{15}	5.4×10^{16}

Table 1: Cluster-prime Inside–Outside cost at $N = 10^6$, $L = 16$.

3.3 Speedup

$$\frac{C_{\text{total}}^{\text{full}}}{C_{\text{total}}^{\text{cluster}}} = \frac{V^3}{K^3} = \left(\frac{V}{K}\right)^3. \quad (14)$$

At $V = 1024$, $K = 16$:

$$\text{speedup} = 64^3 = 262,144 \times. \quad (15)$$

3.4 Memory

Rule tensor: $K^3 \cdot 4 = 16$ KB per level, 256 KB total. Inside table: $NLK \cdot 4 = 1$ GB. Both fit comfortably.

Proposition 2 (Cluster-prime tractability). *At $K = 16$, $N = 10^6$, $L = 16$, cluster-prime Inside–Outside EM requires $\sim 10^{13}$ operations over 50 sweeps and < 2 GB memory. It runs in hours on a single A100 and minutes on a small cluster.*

4 Comparison with QKV Transformer Training

4.1 Transformer cost at matched scale

For a Transformer with L_{tf} layers, model dimension d , vocabulary V , and context window C :

$$C_{\text{QKV}} = 3d^2, \quad (16)$$

$$C_{\text{attn}} = 2Cd, \quad (17)$$

$$C_{\text{FFN}} = 8d^2. \quad (18)$$

Per token per layer: $C_{\text{layer}} = 3d^2 + 2Cd + 8d^2 = 11d^2 + 2Cd$.

Full context ($C = N = 10^6$). With $d = 512$, $L_{\text{tf}} = 8$:

$$C_{\text{fwd}} = NL_{\text{tf}}(11d^2 + 2Nd) \approx 8.2 \times 10^{15} \text{ MACs}. \quad (19)$$

Training step $\approx 3 \times$ forward: 2.5×10^{16} MACs. Over 48,000 steps:

$$C_{\text{train}}^{\text{full}} \approx 1.2 \times 10^{21} \text{ MACs}. \quad (20)$$

Windowed context ($C = 4096$). Per token per layer: $11d^2 + 2 \cdot 4096 \cdot 512 \approx 7.1 \times 10^6$. Total over 48,000 steps:

$$C_{\text{train}}^{\text{window}} \approx 1.4 \times 10^{19} \text{ MACs.} \quad (21)$$

4.2 Comparison table

Method	Total ops	Time ($1 \times \text{A100}$)	Hierarchy detected?
Transformer, full context, 48k steps	1.2×10^{21}	~ 12 days	No
Transformer, windowed, 48k steps	1.4×10^{19}	~ 12 hours	No
Cluster-prime EM, $K = 16$, 50 sweeps	1.3×10^{13}	~ 32 seconds	Yes, depth 16

Table 2: Compute comparison at $N = 10^6$, $V = 1024$.

4.3 Architectural argument

Proposition 3 (Bilinear attention ceiling). *A single attention head computes $\text{Softmax}(QK^\top / \sqrt{d_k})V$, which is a rank-2 bilinear form followed by a linear map. Stacking h heads yields a sum of h rank-2 terms, i.e. a CP approximation of rank at most h . Matching a rank- m rule tensor exactly requires $h \geq V^{m-1}$ heads.*

Proof. The rule tensor $P_l(p \rightarrow x_1, \dots, x_m)$ has m child indices. A rank-2 bilinear form contracts only two indices at a time. To represent an m -index tensor exactly as a sum of rank-2 terms, one needs at least V^{m-1} such terms (the CP rank of a generic m -way tensor of dimension V is $\Theta(V^{m-1})$). Each head contributes one rank-2 term, so $h \geq V^{m-1}$. $\square$

Corollary 1. *At $V = 1024$ and $m = 3$ (binary grammar), exact representation requires $\geq 10^6$ heads. At $m = 4$ (ternary grammar), $\geq 10^9$ heads. This is the architectural wall.*

5 Rank-4+ Tensor Contractions

5.1 Ternary grammars

A ternary grammar has rules $P_l(p \rightarrow x, y, z) \in \mathbb{R}^{V \times V \times V \times V}$, a rank-4 tensor. The Inside recursion becomes

$$T_l(b, n, p) = \sum_{x, y, z \in V} P_l(p \rightarrow x, y, z) \cdot T_{l-1}(b, 3n, x) \cdot T_{l-1}(b, 3n+1, y) \cdot T_{l-1}(b, 3n+2, z). \quad (22)$$

5.2 Cost ladder

For an m -ary grammar (rank- $(m+1)$ rule tensor):

$$C_{\text{node}}^{\text{full}}(m) = 3V^{m+1}, \quad (23)$$

$$C_{\text{node}}^{\text{cluster}}(m) = 3K^{m+1}. \quad (24)$$

Total over 50 sweeps:

$$C_{\text{total}}^{\text{cluster}}(m) \approx 4 \cdot 50 \cdot NLK^{m+1} = 200NLK^{m+1}. \quad (25)$$

Arity m	Tensor order	Total ops (50 sweeps, $K = 16$)	Feasibility
2 (binary)	Rank-3	1.3×10^{13}	Seconds
3 (ternary)	Rank-4	2.1×10^{14}	Minutes
	4	3.4×10^{15}	Hours
	5	5.4×10^{16}	Day
	6	8.6×10^{17}	Borderline
	7	1.4×10^{19}	Imploding

Table 3: Rank ladder at $N = 10^6$, $L = 16$, $K = 16$.

5.3 Head requirement vs. cluster-prime cost

Tensor order	Heads needed (exact)	Cluster-prime ops ($K = 16$)
Rank-3	$V = 1024$	$K^3 = 4,096$
Rank-4	$V^2 = 10^6$	$K^4 = 65,536$
Rank-5	$V^3 = 10^9$	$K^5 = 10^6$
Rank-6	$V^4 = 10^{12}$	$K^6 = 1.7 \times 10^7$

Table 4: Transformer head count vs. cluster-prime cost for matched tensor order.

Corollary 2 (Compression buys tensor orders). *The cluster-prime compression reduces the per-node cost from V^{m+1} to K^{m+1} , a factor of $(V/K)^{m+1}$. At $V/K = 64$, each additional tensor order costs a factor of 64 in the compressed regime but a factor of 1024 in the full regime. The compression therefore buys $\log_{64}(1024) \approx 1.67$ extra tensor orders of headroom per unit of compute.*

6 Berkovich–Hodge Formulation

6.1 Balls and the ultrametric tree

In the p -adic setting, the expectation of a random variable is a ball $B(a, r) = \{x : |x - a|_p \leq r\}$, not a point. Balls satisfy the ultrametric dichotomy:

$$B(a, r) \cap B(b, s) \in \{\emptyset, B(a, r), B(b, s)\}. \quad (26)$$

The space of balls carries a tree structure — the Berkovich space $\mathcal{B}$ — with skeleton Σ_r at resolution r .

6.2 Cluster primes as balls

Definition 1 (Cluster prime). *A cluster prime at resolution r is a vertex of the skeleton Σ_r that carries positive probability mass. The set of cluster primes at resolution r is $C_r = \{c \in \Sigma_r : \mu(B_c) > 0\}$.*

The number of cluster primes $K = |C_r|$ is the effective alphabet at resolution r . Varying r moves along the skeleton and changes K .

6.3 Tropical divergence

Replace the Kullback–Leibler divergence with its tropical analogue:

$$D_{\text{trop}}(\mu\|\nu) = \sum_{B \in \mathcal{B}} \mu(B) \cdot v_p\left(\frac{\mu(B)}{\nu(B)}\right), \quad (27)$$

where $v_p = v_p$ is the p -adic valuation. This divergence is piecewise linear on the skeleton Σ_r and satisfies a tropical Pythagorean identity:

$$D_{\text{trop}}(\mu\|\nu) = D_{\text{trop}}(\mu\|\mu^*) + D_{\text{trop}}(\mu^*\|\nu), \quad (28)$$

where μ^* is the m -projection of μ onto the model manifold at resolution r .

6.4 Filtration and spectral sequence

The nesting of balls across depths,

$$B(a_0, r_0) \supset B(a_1, r_1) \supset \cdots \supset B(a_L, r_L), \quad (29)$$

is a decreasing filtration of the parameter space. The spectral sequence of this filtration has pages $E_1, \dots, E_L$, with differentials d_k given by the Inside contractions at level k .

Proposition 4 (EM as spectral sequence degeneration). *The cluster-prime Inside–Outside EM is the degeneration of the spectral sequence associated to the ball filtration. The E_1 page is the level-1 cluster primes; the E_L page is the deep hierarchical state. The E -step is the m -projection onto the skeleton; the M -step is the e -projection. Monotonicity of the tropical energy replaces monotonicity of the log-likelihood.*

6.5 Hodge–Tate weights as mass concentration rates

The Hodge–Tate weights of the filtered cluster-prime space are the slopes of probability mass concentration per level:

$$w_l = \frac{d \log \mu(B_l)}{dl}. \quad (30)$$

These weights measure the effective rank at each depth and determine when the spectral sequence degenerates (i.e., when no new cluster primes appear).

6.6 Amortization via the filtration

The critical computational role of the Hodge filtration is amortization. If cluster primes must be recomputed per node, the clustering cost is $O(NV^3)$, which implodes. If cluster primes are shared hierarchically via the filtration — coarse primes at shallow levels, refined primes at deep levels — the clustering cost drops to $O(NLK^3)$, matching the Inside cost.

Proposition 5 (Amortization). *Let the filtration $F^\bullet$ refine cluster primes from level l to level $l+1$ with refinement factor $\rho \leq K$. Then the amortized clustering cost over L levels is $O(NLK^3)$, and the total pipeline cost is dominated by the Inside recursion.*

7 Summary

- (i) Full-symbol Inside–Outside at $V = 1024$, $N = 10^6$, $L = 16$ requires $\sim 10^{18}$ operations and > 128 GB. It implodes.
- (ii) Cluster-prime Inside–Outside at $K = 16$ requires $\sim 10^{13}$ operations and < 2 GB. It runs in hours on one GPU.
- (iii) The speedup is $(V/K)^3 = 262,144\times$ at $K = 16$.
- (iv) Transformer training at matched scale costs 10^{19} – 10^{21} MACs and *still fails* to detect the hierarchy. The cluster-prime method is $10^6\times$ cheaper and succeeds.
- (v) The architectural ceiling of bilinear attention is rank-2 per head. Matching a rank- m rule tensor requires V^{m-1} heads. This is the wall.
- (vi) Cluster-prime compression extends the method to rank-4 (ternary), rank-5, and rank-6 grammars at $K = 16$, with costs 10^{14} , 10^{15} , 10^{16} operations over 50 sweeps respectively.
- (vii) The Berkovich–Hodge formulation identifies the resolution radius r as the amortization knob. The filtration is not decorative: it is the mechanism that keeps cluster-prime discovery tractable.
- (viii) The tropical divergence (27) replaces KL divergence, with a piecewise-linear Pythagorean identity on the skeleton.